

Course Syllabus

ECE/ENGRD 2720. Data Science for Engineers

Spring 2026.

1. Course Information

Lectures:	Monday and Wednesday 2-55 to 4:10 pm
Discussion:	Thu and Friday (problem solving sessions with TAs)
Pre-requisites:	MATH 1920 and either CS 1110/CS 1112, MATH 2940 (coreq) Students need to install Jupyter notebook on their laptop to run Python.
Instructor:	Prof. Vikram Krishnamurthy vikramk@cornell.edu
Course Material:	Detailed slides are provided along with problem sets and assignments.

NO auditors are allowed for ECE/ENGRD 2720

2. Course Description and Objectives

An introduction to data science for engineers. The data science workflow: acquisition and cleansing, exploration and modeling, prediction and decision making, visualization and presentation. Tools for data science including numerical optimization, the Discrete Fourier Transform, Principal Component Analysis, and probability with a focus on statistical inference and correlation methods. Techniques for different steps in the workflow including outlier detection, filtering, regression, classification, and techniques for avoiding overfitting. Methods for combining domain-agnostic data analysis tools with the types of domain-specific knowledge that are common in engineering. Ethical considerations. Optional topics include classification via neural networks, outlier detection, and Markov chains. Programming projects in Python.

By the end of the course, students should be able to: (i) understand the data science workflow and basic models for data generation; (ii) understand the underlying mathematics, formulate and design algorithms for exploratory data analysis and least squares inference (iii) able to formulate and solve unconstrained optimization and linear programming using Matlab or Python optimization solvers, (iv) Understand and formulate elementary probabilistic models and statistical inference for data.

3. Academic Integrity

Cornell University's Code of Academic Integrity is available at <http://cuinfo.cornell.edu/aic.cfm>

4. Accommodations for Students with Disabilities

In compliance with the Cornell University policy and equal access laws, the instructor is available to discuss appropriate academic accommodations that may be required for students with disabilities. Students are encouraged to register with Student Disability Services to verify their eligibility for appropriate accommodations.